Sashreek Dhaimodkar

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PROFILE

I'm a Computer Science master's student passionate about exploring opportunities across all its field. My experience spans from software development, model training, workflow optimization, to ensuring reliable infrastructure performance. Proficient in Python, C++, SQL and Linux with hands-on experience in cloud environments, scalable model deployment and tackling complex technical problems with a curious and structured mindset. Above all, I'm driven by the challenge of turning innovative AI ideas into practical, scalable solutions and build intelligent systems that combine data, algorithms, and real-world impact.

PROFESSIONAL EXPERIENCE

Student Assistant 03/2026 – Present

Paderborn University (VirtuChem Lab)

- Developing features for **VirtuTutor2**, a Meta Quest 3 VR chemistry lab with an **AI tutoring system** built in Unity (C#), Python and FastAPI.
- Implementing a **RAG-based personalization system** using ChromaDB and sentence transformers to deliver history-aware LLM tutoring responses.

Associate Software Engineer 07/2023 – 08/2024

OneShield Software

- Designed and developed multiple forms and modules for enterprise insurance software using **Python**, **Java**, **XML**, **PL/SQL** and **Proprietary no-code software**.
- Diagnosed and resolved complex system errors, improving system reliability and user experience.
- Conducted daily **system health checks** to ensure uptime and performance stability.
- Integrated and maintained **AWS services** for deployment and monitoring.
- Collaborated cross-functionally with QA and product teams to deliver scalable and secure solutions.
- Worked in an **Agile environment** following **Scrum methodologies**, participating in sprint planning, daily stand-ups, and retrospective meetings to ensure timely and high-quality deliverables.

ML Intern 08/2022 – 10/2022

Persistent Systems

- Contributed to the development of a **Federated Learning Platform** enabling distributed model training.
- Implemented the **Flower** framework for federated learning workflows across multiple nodes.
- Logged and monitored ML experiments using **MLflow**, improving model versioning and reproducibility.
- Containerized applications using **Docker** to streamline deployment and testing workflows.
- Built interactive web interfaces for ML models using **Streamlit** and **FastAPI**.
- **Deployed the complete application on AWS**, leveraging cloud infrastructure for scalable and reliable performance.

EDUCATION

Computer Science Master of Science 10/2024 – Present

Universität Paderborn

Computer Engineering 06/2019 – 07/2023

Padre Conceicao College of Engineering

Bachelor of Engineering, 9.34CGPA (1.33 in German system)

SKILLS

Python/ C++/ SQL/ Docker/ FastAPI/ Streamlit/ MLflow/ AWS/ HPC/ Slurm/ Tensorflow/ PyTorch/ Federated Learning/ LLM Training/ Data Preprocessing/ MLOps/ Linux Shell/ Git/ JIRA/ Agile & Scrum/ Analytical Skills/ Machine Learning/ Deep Learning/ High level knowledge of Prompt Engineering

PROJECTS

Project Group - Training Multilingual Large Language Model 04/2025 – Present
Data Science Group, Paderborn University

- Working on development and optimization of a **large-scale multilingual language model**, focusing on scalability, efficiency, and low-resource language generalization.
- Designed and implemented **fine-tuning pipelines** using **cross-lingual transfer learning**, **LoRA/QLoRA adapters**, and experimental **XLoRA mixtures** for adaptive training across language families.
- Executing large-scale distributed training on the **PC2 OTUS High-Performance Computing (HPC) cluster** and virtual machines, leveraging **NVIDIA H100 GPUs**.
- Developed and optimized **training scripts, data preprocessing pipelines, and monitoring utilities** to ensure fault-tolerant and reproducible experimentation.

Synthetic MRI Genration 10/2022 – 06/2023
Final Year Project - Predicting MRI Progression in Alzheimer's Disease Patients

- Developed a **deep learning model** to simulate MRI progression for Alzheimer's Disease patients, supporting early diagnosis and treatment planning.
- Customized and optimized a **U-Net architecture** for image generation and segmentation tasks.
- Built an **interactive user interface** for model inference and visualization to enhance accessibility for non-technical users.

Project Management System 08/2022 – 08/2022
Mini Project

- Developed windows form platform using C# in Visual Studio.
- Designed login, register, add and delete flows.
- Database integration done using oracle database in Visual Studio.

Sentiment Analysis on Facebook Comments 06/2022
Mini Project

- Developed a **Python-based sentiment analysis application** to classify Facebook comments into positive, negative, or neutral categories.
- Implemented **text preprocessing**, **tokenization**, and **feature extraction** using **NLTK** and **Pandas**.
- Delivered insights on social sentiment trends through data-driven analysis and visualization.

CERTIFICATIONS

Getting started with Python
Coursera

Joy of Computing using Python
NPTEL

Data Analytics with Python
NPTEL

LANGUAGES

English
C1 - IELTS Band 8.0

German
A2 - Grade 2.0